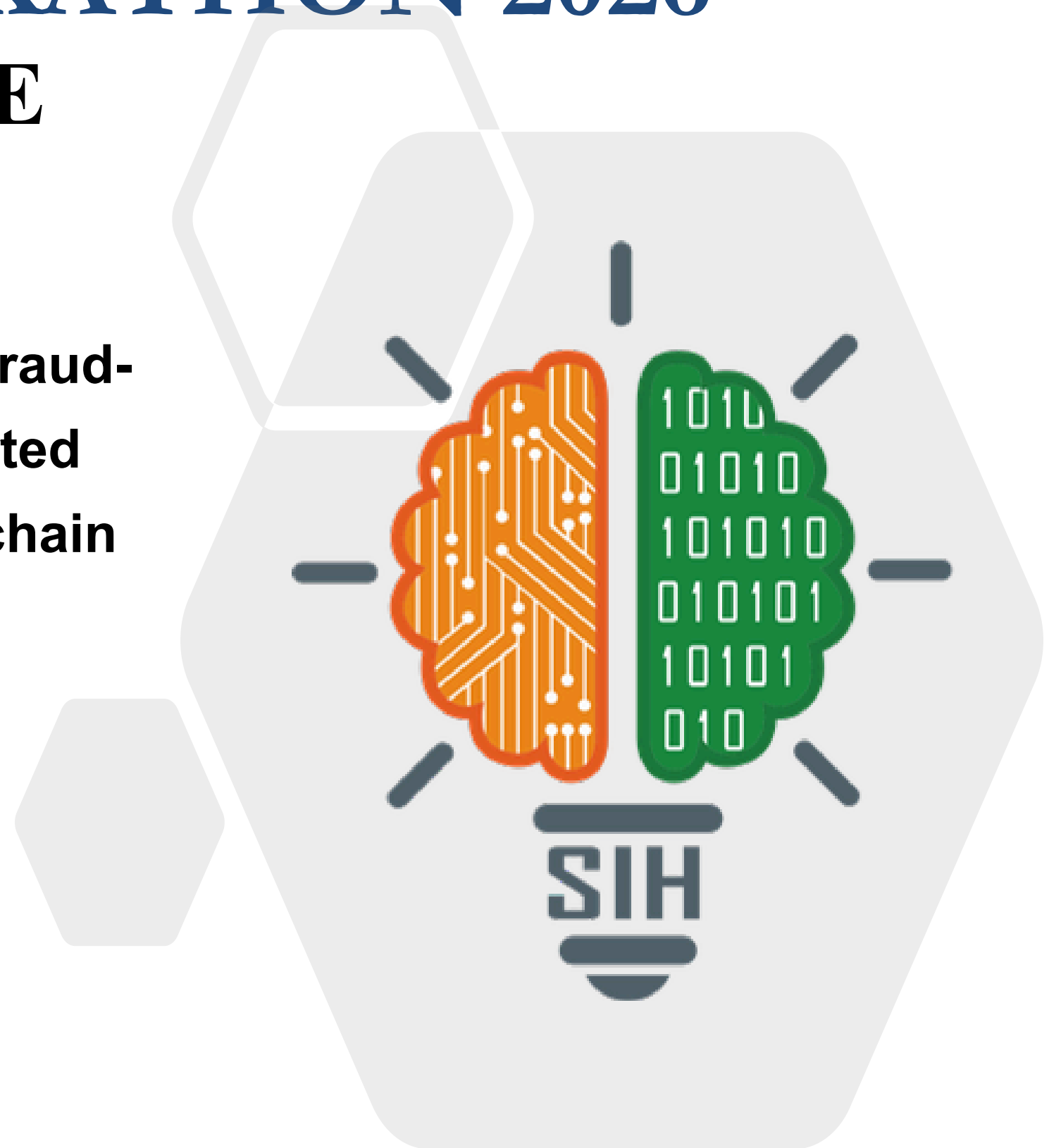


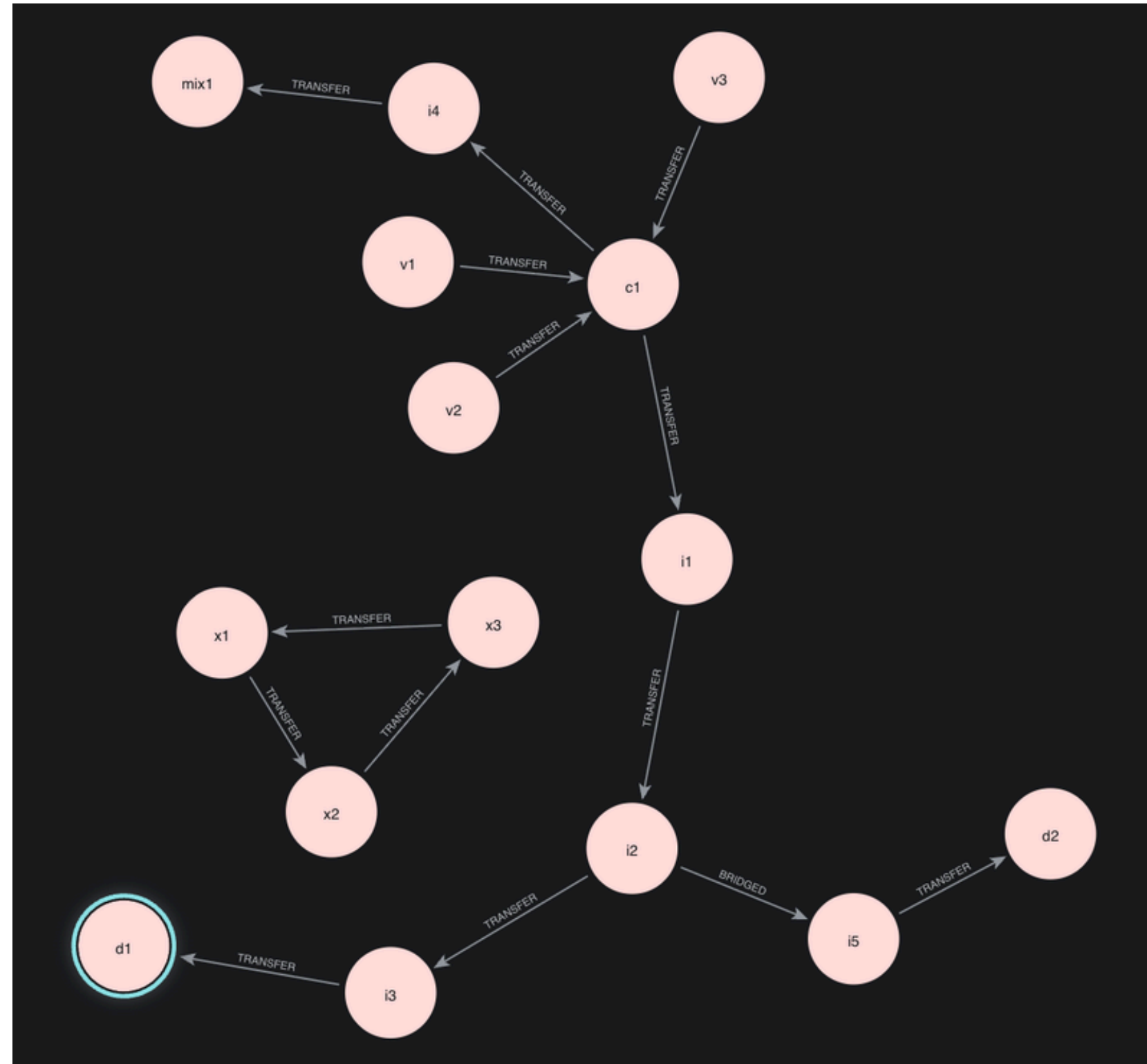
SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID- SIH26183
- Problem Statement Title- Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- 189703
- Team Name (Registered on portal)- Orion_1



ORION: TRACING FRAUD CRYPTO TO THE EXCHANGE, IN REAL TIME



PROPOSED SOLUTION

01 | COMPLAINT-TO-WALLET-TRAIL

Each reported wallet becomes a live multi-chain graph of transfers, bridges and mixers, showing where funds moved and how close they are to an exchange.

02 | LAUNDERING WALLET DETECTION

Typology rules (fan-in, layering, peel chain, mixer/bridge hop, burner, circular flow) plus OddBall and burst scoring separate intermediaries from endpoints and assign each wallet a risk category.

03 | EXCHANGE / VASP ATTRIBUTION

Wallet clustering and intelligence-API labels identify the nearest exchange/VASP receiving a direct deposit, with a confidence score. Models run as a ladder (heuristics → clustering → graph-aware XGBoost), and each must beat the one below.

04 | ACT BEFORE FUNDS EXIT

Fund-flow visualisation, explainable alerts, a min-cut recommender that names the wallets and deposits to freeze, and an auto-generated investigation report.

COMPLAINT → WALLET TRACE → EXCHANGE ATTRIBUTION → FREEZE & RECOVER

TECH STACK



Backend and Alerts



Scikit-Learn, XGBoost,



React + Cytoscape.js/D3 +
Tailwind for Dashboard

1. DATA SOURCES

- NCRP / SAHYOG complaints
- Blockchain node/indexer APIs
- Exchange/VASP wallet label database
- Mixer/bridge and threat-intel address lists
- Historical fraud-wallet cases

2. PREPROCESSING

- Cleaning
- Multi-chain address normalisation
- Multi-chain transaction indexing
- Time-Window Bucketing
- Multi-chain transaction indexing

3. FUND TRACE & LAUNDERING DETECTION

- Multi-hop, cross-chain trail tracing
- Typology Rules (Cypher)
- OddBall Anomaly Detection
- Temporal Burst scoring.

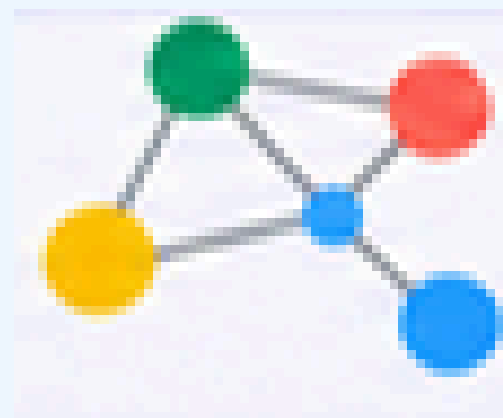
5. ATTRIBUTION & OPTIMISATION

- Exchange wallet clustering
- Graph-aware XGBoost (wallet risk score)
- min-cut fund-blocking
- adaptive adversary stress-test

6. OUTPUT

- Fund-flow graph dashboard
- Real-Time explainable alerts (SMS/email/API)
- Law enforcement interface (SAHYOG/NCRP API)
- Freeze-list + investigation report

4. MONEY-FLOW GRAPH



Entities: Wallet, Exchange/VASP, Transaction

Relationships :
Explicit | Inferred |
Attributed exchange deposit

Challenge / Risk	Risk Level	Mitigation Strategy
No labelled Indian fraud-wallet ground truth	HIGH	Public labelled blockchain datasets + simulated scam typologies (investment, task fraud, sextortion). Production ingests NCRP/SAHYOG complaints via I4C.
Model memorises its own synthetic data	HIGH	Time-based splits, multiple scenarios, mandatory baseline, precision@K.
Wrong attribution of an exchange or wallet owner	HIGH	Confidence score on every attribution, source transactions stored, human verifies before any freeze request.
Alert fatigue and false positives at VASPs/investigators	MEDIUM	Tunable thresholds, explained alerts, reported precision and alerts/day.
Mixers, bridges, privacy coins and chain-hopping	MEDIUM	Cross-chain bridge tracing, adaptive adversary module, low-confidence flag and

Working today: Neo4j graph engine, typology detectors, OddBall + burst scoring (reused from Orion core); wallet-tracing module in progress.

TARGET AUDIANCE



State police and district cyber cells
(faster wallet-to-exchange tracing)



Citizens and victims (better recovery odds)



Exchanges/VASPs and banks (faster freeze requests)



I4C / NCRP / SAHYOG (cross-state intelligence)

KEY IMPACT



Instant exchange attribution from reported wallet



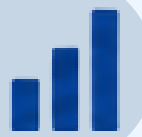
Fewer cold complaints



Freeze within the golden hour



Higher recovery



Cross-jurisdiction intelligence



Coordinated action



Explained alerts, less noise



Faster decisions

Existing Approach

Manual blockchain tracing needs expertise and days
Single-chain explorer lookups
Static accuracy on one dataset

What Orion Adds

Automated real-time tracing to the nearest exchange/VASP
Multi-chain, cross-bridge fund-flow attribution
Baseline-beating, time-split and adversary-tested results

Academic Research Papers

- Akoglu, McGlohon & Faloutsos (2010), "OddBall: Spotting Anomalies in Weighted Graphs," PAKDD - foundation for Tier 2 egonet anomaly scoring
- Weber et al. (2019), graph convolutional networks for Bitcoin AML (Elliptic dataset)
- Meiklejohn et al. (2013), "A Fistful of Bitcoins" (wallet clustering)
- Möser et al. (2018), traceability in the Monero blockchain (privacy-coin limits)

Legal, Industry & Regulatory References

- I4C / NCRP / SAHYOG: complaint data and the wallet-freezing workflow Orion plugs into
- FATF virtual asset/VASP guidance: layering and laundering typologies, the basis of our detectors
- PMLA 2002 (VDA service providers): basis for VASP obligations and freeze cooperation
- BNSS 2023 and BSA 2023: legal basis for freezing suspect funds and admissibility of electronic records, so every alert stores its source transactions
- DPDP Act 2023: lawful data handling, retention limits and role-based access

Important Links :-

- **Project Report:** <https://drive.google.com/file/d/1AghmjYd16m16lIfYuZcn94XoD6Y5DCr9/view?usp=sharing>